

ECE 222 Homework #4

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Homework #4 only has five questions, and each question is worth 2 points.

The solutions for Homework #4 have been provided on Canvas to help you prepare for Midterm 1. Please try to solve the questions on your own before looking at the solutions.

Electrical Properties of Materials

Do the following problems:

- a) From the textbook: 4.41, 4.46**
- b) 3 additional problems from other books/exams (scanned below)**

- 4.41 A cylindrical bar of silicon has a radius of 4 mm and a length of 8 cm. If a voltage of 5 V is applied between the ends of the bar and $\mu_e = 0.13 \text{ (m}^2/\text{V} \cdot \text{s)}$, $\mu_h = 0.05 \text{ (m}^2/\text{V} \cdot \text{s)}$, $N_e = 1.5 \times 10^{16} \text{ electrons/m}^3$, and $N_h = N_e$, find the following:
- the conductivity of silicon,
 - the current I flowing in the bar,
 - * the drift velocities $\mathbf{u}_e$ and $\mathbf{u}_h$,
 - the resistance of the bar, and
 - the power dissipated in the bar.
- 4.46 A $2 \times 10^{-3} \text{ mm}$ thick square sheet of aluminum has $5 \text{ cm} \times 5 \text{ cm}$ faces. Find the following:
- The resistance between opposite edges on a square face.
 - The resistance between the two square faces. (See Appendix B for the electrical constants of materials.)

Table B-2 Conductivity σ of Some Common Materials^a

Material	Conductivity σ (S/m)	Material	Conductivity σ (S/m)
Conductors		Semiconductors	
Silver	6.2×10^7	Pure germanium	2.2
Copper	5.8×10^7	Pure silicon	4.4×10^{-4}
Gold	4.1×10^7	Insulators	
Aluminum	3.5×10^7	Wet soil	$\sim 10^{-2}$
Tungsten	1.8×10^7	Fresh water	$\sim 10^{-3}$
Zinc	1.7×10^7	Distilled water	$\sim 10^{-4}$
Brass	1.5×10^7	Dry soil	$\sim 10^{-4}$
Iron	10^7	Glass	10^{-12}
Bronze	10^7	Hard rubber	10^{-15}
Tin	9×10^6	Paraffin	10^{-15}
Lead	5×10^6	Mica	10^{-15}
Mercury	10^6	Fused quartz	10^{-17}
Carbon	3×10^4	Wax	10^{-17}
Seawater	4		
Animal body (average)	0.3 (poor cond.)		

^aThese are low-frequency values at room temperature (20 °C).

Three additional problems

- 5.22** In a dielectric material ($\epsilon = 5\epsilon_0$), the potential field $V = 10x^2yz - 5z^2$ V, determine (a) $\mathbf{E}$, (b) $\mathbf{D}$, (c) $\mathbf{P}$, (d) ρ_v .
- 5.25** Two point charges in free space are separated by distance d and exert a force 2.6 nN on each other. The force becomes 1.5 nN when the free space is replaced by a homogeneous dielectric material. Calculate the dielectric constant of the material.

Problem 5:

Charge Q has been placed on a metallic sphere of radius a (i.e., the sphere is made of an excellent conductor). The surrounding medium is air (relative permittivity $\epsilon_r = 1$).

- a) Use Gauss's law to calculate the electric flux density $\vec{D}$ and electric field intensity $\vec{E}$ everywhere in space. Graph $D(R)$ vs R and $\epsilon_0 E(R)$ vs R . (Hint: The sphere is metallic. That should guide your assumptions about how the charge is distributed.)
- b) A thick dielectric spherical shell is added to the system (i.e., in addition to the charged sphere from before). The dielectric extends from $R = a$ to $R = b$, and has relative permittivity of $\epsilon_r = 2$. Calculate and graph $D(R)$ vs R , $\epsilon_0 E(R)$ vs R , and $P(R)$ vs R .
- c) Calculate the potential difference $V(a) - V(b)$ for the setup described in part (b) (i.e., both the conductor and the dielectric shell together).

Solutions to starred problems:

4.41 $\square$ (c) $\mathbf{u}_e = -8.125\mathbf{E}/|\mathbf{E}|$ (m/s); $\mathbf{u}_h = 3.125\mathbf{E}/|\mathbf{E}|$ (m/s)

4.45 $\square$ $R = 4.2$ (m Ω)

6.20 $\square$ $\rho_v = (8y/\omega) \sin \omega t + C_0$, where C_0 is a constant of integration.